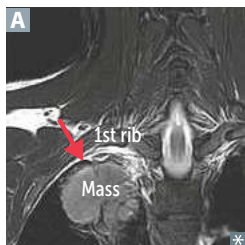
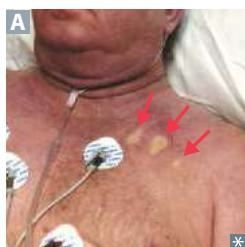


Pancoast tumor

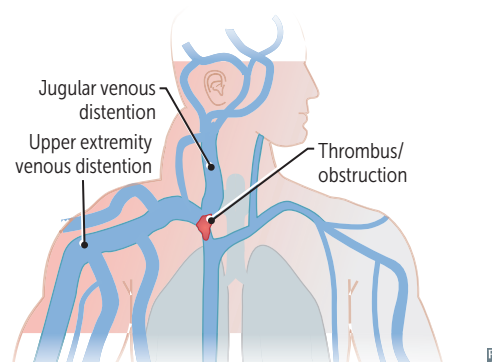
Also called superior sulcus tumor. Carcinoma (most commonly NSCLC) that occurs in the apex of lung **A** may cause Pancoast syndrome by invading/compressing local structures.

Compression of locoregional structures may cause array of findings:

- Recurrent laryngeal nerve → hoarseness
- Stellate ganglion → Horner syndrome (ipsilateral ptosis, miosis, anhidrosis)
- Superior vena cava → SVC syndrome
- Brachiocephalic vein → brachiocephalic syndrome (unilateral symptoms)
- Brachial plexus → shoulder pain, sensorimotor deficits (eg, atrophy of intrinsic muscles of the hand)
- Phrenic nerve → hemidiaphragm paralysis (hemidiaphragm elevation on CXR)

Superior vena cava syndrome

Obstruction of the SVC (eg, thrombus, tumor) impairs blood drainage from the head (“facial plethora”; note blanching after fingertip pressure in **A**), neck (jugular venous distention, laryngeal/pharyngeal edema), and upper extremities (edema). Commonly caused by malignancy (eg, mediastinal mass, Pancoast tumor) and thrombosis from indwelling catheters. Medical emergency. Can raise intracranial pressure (if obstruction is severe) → headaches, dizziness, ↑ risk of aneurysm/rupture of intracranial arteries.

**► RESPIRATORY—PHARMACOLOGY****Asthma drugs**

Bronchoconstriction is mediated by (1) inflammatory processes and (2) parasympathetic tone; therapy is directed at these 2 pathways.

Inhaled β_2 -agonists

Albuterol (short-acting), salmeterol, formoterol—relax bronchial smooth muscle. Can cause tremor, arrhythmia. Albuterol is for acute symptoms.

Inhaled or oral glucocorticoids

Fluticasone, budesonide—inhibit the synthesis of virtually all cytokines. Inactivate NF- κ B, the transcription factor that induces production of TNF- α and other inflammatory agents. 1st-line therapy for chronic asthma. Use a spacer or rinse mouth after use to prevent oral thrush.

Muscarinic antagonists

Tiotropium, ipratropium—competitively block muscarinic receptors, preventing bronchoconstriction. Also used for COPD. Tiotropium is long acting.

Antileukotrienes

Montelukast, zafirlukast—block leukotriene receptors (CysLT1). Especially good for aspirin-induced and exercise-induced asthma.

Zileuton—5-lipoxygenase inhibitor. ↓ conversion of arachidonic acid to leukotrienes. Hepatotoxic.

Anti-IgE monoclonal therapy

Omalizumab—binds mostly unbound serum IgE and blocks binding to Fc ϵ RI. Used in allergic asthma with ↑ IgE levels resistant to inhaled glucocorticoids and long-acting β_2 -agonists.

Methylxanthines

Theophylline—likely causes bronchodilation by inhibiting phosphodiesterase → ↑ cAMP levels due to ↓ cAMP hydrolysis. Limited use due to narrow therapeutic index (cardiotoxicity, neurotoxicity); metabolized by cytochrome P-450. Blocks actions of adenosine.

PDE-4 Inhibitors

Roflumilast—inhibits phosphodiesterase → ↑ cAMP → bronchodilation, ↓ airway inflammation. Used in COPD to reduce exacerbations.

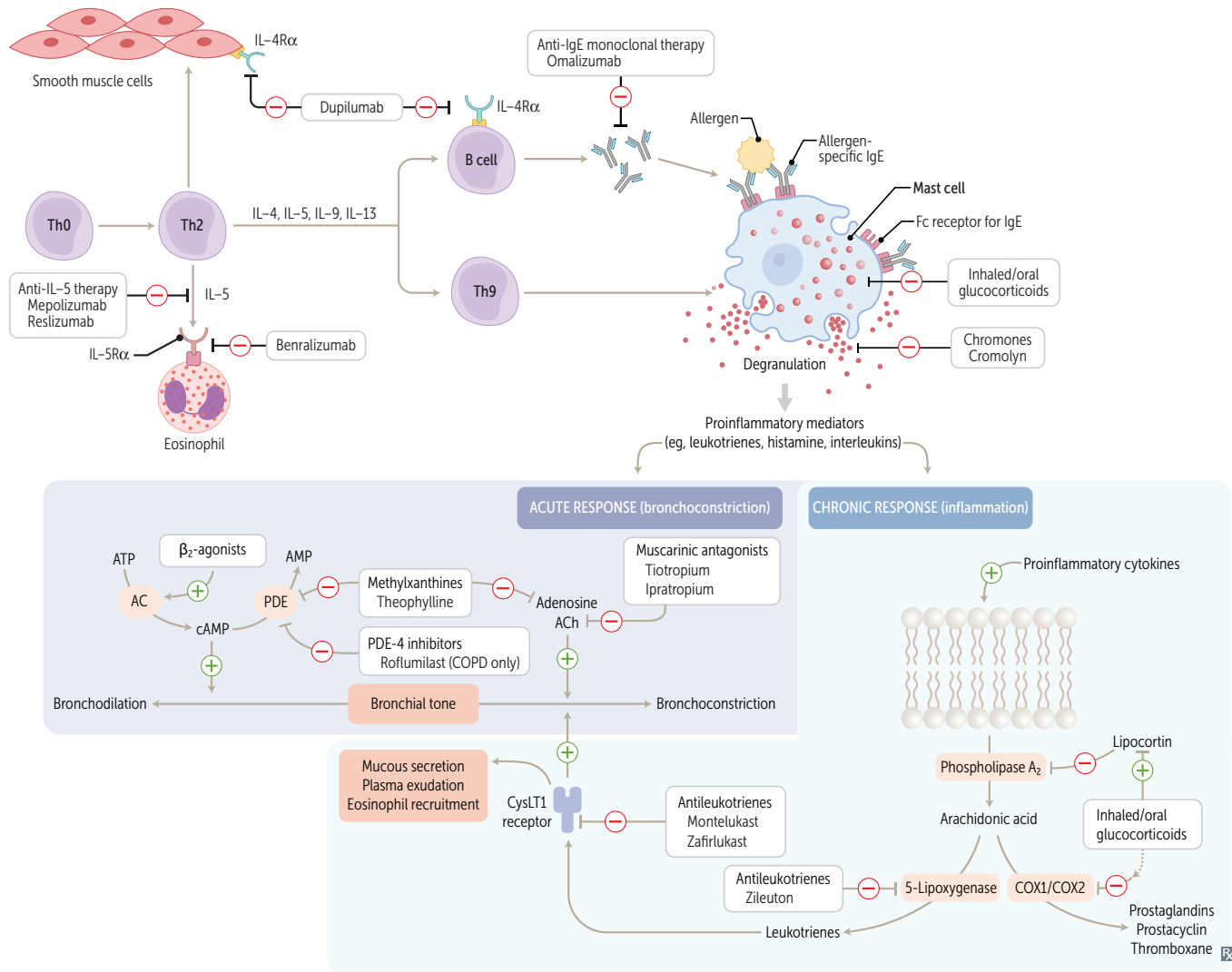
Chromones

Cromolyn—prevents mast cell degranulation. Prevents acute asthma symptoms. Rarely used.

Asthma drugs (continued)**Biologics**

Mepolizumab, reslizumab—against IL-5. **Benralizumab**—against IL-5 receptor α . Prevent eosinophil differentiation, maturation, activation, and survival mediated by IL-5 stimulation. For maintenance therapy in severe eosinophilic asthma.

Dupilumab—against IL-4 and IL-13.

**H₁-blockers**

Also called antihistamines. Reversible inhibitors of H₁ histamine receptors (inverse agonists).

First generation

Diphenhydramine, dimenhydrinate, chlorpheniramine, doxylamine. Names usually contain “-en/-ine” or “-en/-ate.”

CLINICAL USE

Allergy, motion sickness, vomiting in pregnancy, sleep aid.

ADVERSE EFFECTS

Sedation, antimuscarinic, anti- α -adrenergic.

Second generation

Loratadine, fexofenadine, desloratadine, cetirizine. Names usually end in “-adine.” Setirizine (cetirizine) is second-generation agent.

CLINICAL USE

Allergy.

ADVERSE EFFECTS

Far less sedating than 1st generation because of ↓ entry into CNS.

Dextromethorphan

Antitussive (antagonizes NMDA glutamate receptors can act as a hallucinogenic dissociative agent similar to ketamine at high doses (and may be combined with bupropion as a fast acting antidepressant). Synthetic codeine analog. Has mild opioid effect when used in excess. Naloxone can be given for overdose. Mild abuse potential. May cause serotonin syndrome if combined with other serotonergic agents.

Pseudoephedrine, phenylephrine

MECHANISM	Activation of α -adrenergic receptors in nasal mucosa $\rightarrow$ local vasoconstriction.
CLINICAL USE	Reduce hyperemia, edema (used as nasal decongestants); open obstructed eustachian tubes.
ADVERSE EFFECTS	Hypertension. Rebound congestion (rhinitis medicamentosa) if used more than 4–6 days. Associated with tachyphylaxis. Can also cause CNS stimulation/anxiety (pseudoephedrine).

Pulmonary hypertension drugs

DRUG	MECHANISM	CLINICAL NOTES
Endothelin receptor antagonists	Competitively antagonizes endothelin-1 receptors $\rightarrow$ $\downarrow$ pulmonary vascular resistance.	Hepatotoxic (monitor LFTs). Example: bosentan.
PDE-5 inhibitors	Inhibits PDE-5 $\rightarrow$ $\uparrow$ cGMP $\rightarrow$ prolonged vasodilatory effect of NO.	Also used to treat erectile dysfunction. Contraindicated when taking nitroglycerin or other nitrates (due to risk of severe hypotension). Example: sildenafil.
Prostacyclin analogs	PGI ₂ (prostacyclin) with direct vasodilatory effects on pulmonary and systemic arterial vascular beds. Inhibits platelet aggregation.	Adverse effects: flushing, jaw pain. Examples: epoprostenol, iloprost.

